CO2 AT3

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**14

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour
Total Marks:30

QUESTIONS: 3

Annexure A

Dry Run Evaluation

Code of Conduct:

I SRINIVASAN.A 192411025 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A. Srini

20+26+18+18+8

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Q3. Uniform Cost Search (UCS).

Uniform cost search (UCS) expands the node with lowest path cost from the starting node.

Start = A, Goal = D

Priority Queue Trace

Step	Priority Queue	Expanded	Cost
1.	B(2), E(4)	A	0
2.	E(3), D(7)	B	2
3.	C(5), D(6)	E	3
4.	D(6)	C	5
5.	—	D	6

Explanation:

1. Initially, A is expanded. B has cost 2 and E has cost 4.
2. B is selected because it has the lowest cost. Through B, E gets a better cost of 3.
3. E is expanded. It reaches C with cost 5 and D with cost 6.
4. C is expanded, but it does not provide a better path to D.

5. D is Selected with cost 6, so the goal is reached

Least - cost path

$\boxed{A \rightarrow B \rightarrow F \rightarrow D}$

Total cost

$2+1+3 = \boxed{6}$

Conclusion: UCS finds the optimal path from A to D with a minimum cost of 6.

Q4. Problem formulation.

The AI system has to find a route from the college Gate to the CSE laboratory.

problem components

1. Initial state:
college Gate
2. Goal state:
CSE laboratory
3. Actions/operators:
The AI can move between connected locations. Such as:
* Gate → Block A
* Block A → Library
* Library → Block B
* Block B → CSE LAB
* Move through Canteen as an alternate route.

search state Space :

{ Gate, Block A, Library, Block B, Canteen, CSE Lab }

These represent the possible Location of the agent

5. Path cost :

Path cost is the total distance, time or effort required to travel from the starting point to the destination.

6. One possible Solution path ;

Gate → Block A → Library → Block B → CSE Lab.

7. Alternate Route :

Gate → Canteen → Block B → CSE Lab.

Conclusion ;

The AI represents the campus as state-space search problem and selects a suitable route based on the path cost.

Q5. Intelligent Agent and Rationality

A smart vaccum robot operates in two rooms, A and B. It can perform move left, Move Right and Suck.

The goal of the robot is to keep both rooms clean.

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Location	Room Status	Rational Action
A	Dirty	Suck
A	clean	Move Right
B	Dirty	Suck
B	clean	Move Left.

Justification

1. A - Dirty:

The robot performs Suck because the room contains dirt.

2. A clean:

The robot performs Move Right to go to Room B and check its condition.

3. B - Dirty:

The robot perform Suck to clean Room B.

4. B - clan:

The robot performs Move left to return to Room A and check it.

Rationality.

The ~~robot~~

Conclusion: The vaccum cleaner is a rational intelligent agent because it uses its current percepts to select the most appropriate action